

# OPEN BY DESIGN Open Science

## IS, DS, AI, and SD

A concise view of principles, infrastructures, social responsibility, and Brazil's strategic agenda.

### Washington Segundo

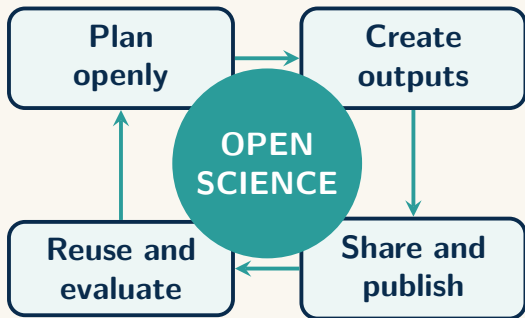
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# Open science redesigns the entire research cycle

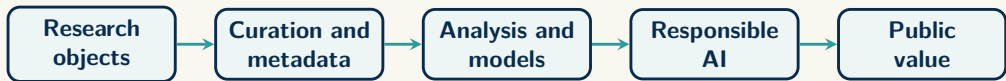


## A PARADIGM SHIFT

- Openness begins at planning, not after publication.
- Publications, data, code, methods, and reviews become reusable outputs.
- Governance, preservation, and participation are part of scientific quality.

**Core idea:** knowledge should be accessible, auditable, interoperable, and useful to society.

# Information Science is the foundation of reliable AI



Data, publications, code Context and provenance Patterns and evidence Grounded automation Decisions and services

## Information Science

Organizes, curates, represents, and  
mediates knowledge

## Data Science

Analyzes patterns, models  
phenomena, and extracts value

## Artificial Intelligence

Automates cognition, prediction, and  
language interaction



# A long evolution led from private exchange to open ecosystems



Digital networks separated the traditional functions of journals: record, certification, dissemination, and preservation.



# Ibict has shaped scientific information since 1954

# 1954

Founded as Brazil's national  
institute dedicated to  
bibliography and scientific  
documentation

## Today

A research unit of the  
Ministry of Science,  
Technology and Innovation  
- MCTI

## A LONG-STANDING PUBLIC MISSION

- Organize, preserve, and disseminate Brazil's scientific and technological information.
- Build national services, standards, and interoperable information infrastructures.
- Connect Brazilian research with regional and international knowledge networks.

**A STRONG TRADITION OF OPENNESS** Ibict has played a sustained leadership role in **Open Access** and **Open Science**, advancing repositories, aggregators, research data, digital preservation, persistent identifiers, and public-interest governance.



# The ecosystem is broader than open access

## Open outputs

Publications, research data, software, methods, and educational resources.

## Open processes

Preprints, transparent peer review, responsible assessment, and reproducible workflows.

## Open infrastructures

Repositories, identifiers, metadata standards, preservation, and research information systems.

## Open participation

Citizen science, knowledge dialogue, inclusion, and public engagement.

Infrastructure connects every pillar and keeps openness sustainable.

# Four principles balance technical value and social responsibility

## FAIR

Make data findable, accessible, interoperable, and reusable for people and machines.

## CARE

Protect collective benefit, authority, responsibility, and ethics in data governance.

## TRUST

Build transparent, responsible, user-focused, sustainable, and technically sound repositories.

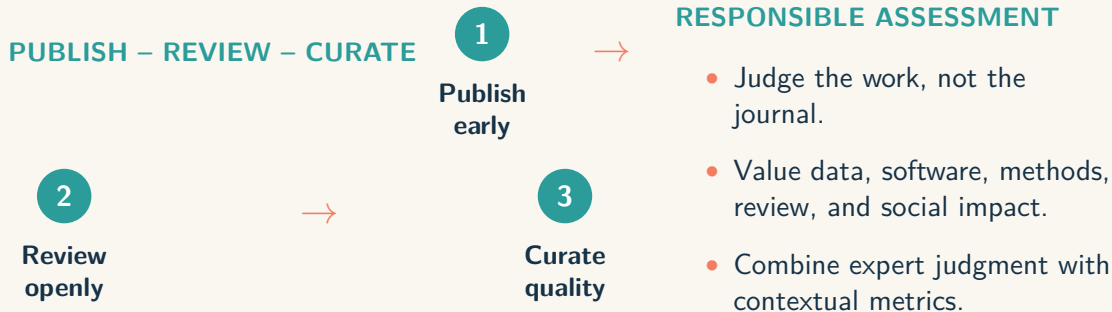
## DEIA

Embed diversity, equity, inclusion, and accessibility in research systems.

**Rule of thumb:** as open as possible, as closed as necessary.



# Communication and assessment are moving toward transparency



Preprints accelerate circulation; open reviews expose reasoning; curation adds context and certification.



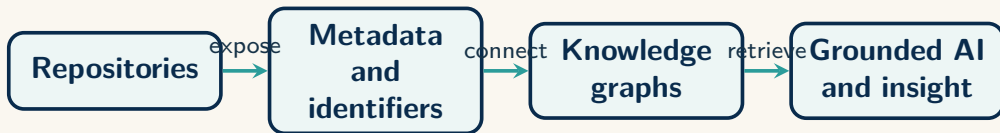
# Open access models distribute cost and power

| Route   | How it opens access                        | Main strength                                             | Main risk                                                    |
|---------|--------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------|
| Green   | Deposit in a repository                    | Uses public infrastructure and can avoid publication fees | Embargoes or restrictive publisher policies                  |
| Gold    | Immediate access on the publisher platform | The version of record is openly available                 | APCs can shift exclusion from readers to authors             |
| Diamond | No fees for readers or authors             | Aligns publishing with the public interest                | Requires durable institutional funding and shared governance |

Publicly funded research creates maximum public value when access, publishing, and governance remain equitable.



# Trustworthy AI begins with trustworthy research infrastructure

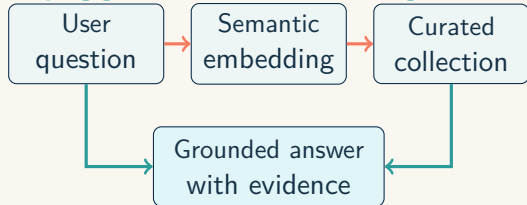


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preservation, governance, standards, and public infrastructure

Preservation and interoperability turn stored objects into reusable evidence.

# Semantic systems connect meaning, not only keywords

## A GROUNDED RETRIEVAL FLOW



## WHAT MAKES IT WORK

- Controlled vocabularies and ontologies reduce ambiguity.
- Multilingual embeddings retrieve concepts across languages.
- RAG grounds generated answers in real documents.
- Rich metadata makes sources traceable and reusable.

# Openness must also deliver informational justice

## THREE DIMENSIONS

- **Distribution:** equitable access to information and infrastructure.
- **Recognition:** respect for diverse people, knowledge, and epistemologies.
- **Participation:** meaningful inclusion in producing and using knowledge.

## ETHICAL SAFEGUARDS

- Consent and risk assessment
- Privacy and controlled access
- Collective and Indigenous data rights
- Protection from harmful or dual use



# Libraries are becoming critical knowledge infrastructure

## AN INSTITUTIONAL EVOLUTION

### 1 Collections

Acquire and provide access

### 2 Digital services

Connect repositories and systems

### 3 Infrastructure

Steward data and long-term access

## THE PROFESSIONAL ROLE EXPANDS

- Data stewardship and metadata enrichment
- Digital preservation and authenticity
- Licensing, privacy, and ethical guidance
- Persistent identifiers and interoperability
- Digital literacy and public mediation
- Coordination across research, IT, government, and society



# Implementation depends on human mediation

## **INCLUSIVE DESIGN** Build for diverse communities

- Accessible interfaces
- Multilingual discovery
- Disciplinary flexibility
- Context-sensitive workflows

## **GOVERNANCE** Define responsibility

- Data control and privacy
- Curation criteria
- Ethical safeguards
- Sustainable ownership

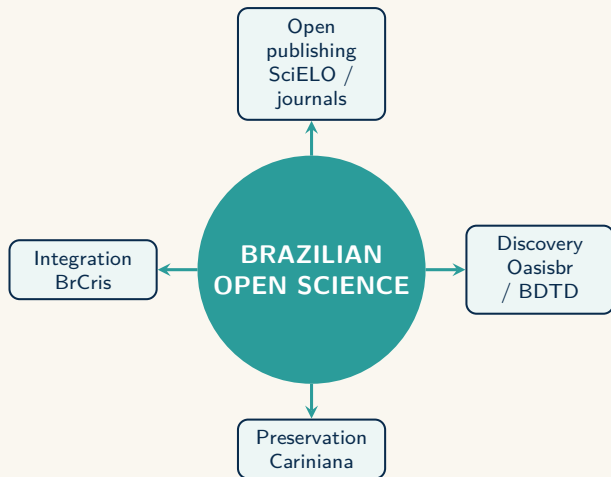
## **DIGITAL LITERACY** Enable critical use

- Practical training
- Limits and uncertainty
- Reproducible practices
- Responsible AI use

Multidisciplinary support teams translate Open Science principles into everyday research practice.



# Brazil already has the foundations of a national ecosystem



## STRATEGIC ASSETS

- SciELO and open journals
- BDTD and Oasisbr
- BrCris and research integration
- Cariniana and digital preservation
- Data, citizen science, and software networks

The next step is coordinated governance, stable funding, and stronger interoperability.

# Ibict connects the national ecosystem in layers

## COLLECT AND PUBLISH

- Institutional repositories and digital libraries
- Brazilian Digital Library of Theses and Dissertations
- Journal and publication support services

## AGGREGATE AND CONNECT

- Oasisbr for unified discovery
- National repository networks and shared standards

## ANALYZE AND GOVERN

- BrCris for research information and policy intelligence
- Laguna for semantic integration and large-scale analysis

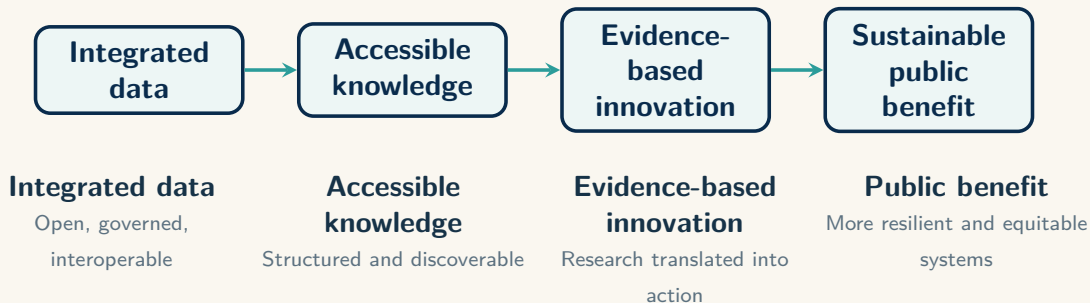
## PRESERVE AND IDENTIFY

- Cariniana for distributed digital preservation
- dARK for persistent, traceable identifiers

Together, these layers preserve institutional autonomy while enabling national-scale



# Scientific infrastructure enables sustainable development



# Three complementary infrastructures strengthen Brazilian research

## BRCRIS Connects the research system

- Researchers and institutions
- Projects and funding
- Outputs and infrastructures

Enables visibility, governance, assessment, and evidence-based policy.

## DARK NETWORK Keeps resources identifiable

- Persistent ARK identifiers
- Traceability and citation
- Long-term access

Protects reproducibility and institutional memory from broken links.

## PROJECT LAGUNA Integrates heterogeneous data

- Semantic enrichment
- Interoperability
- Large-scale discovery

Transforms isolated sources into a shared analytical environment.

# BrCris creates a national view of Brazilian research

## BRAZILIAN CURRENT RESEARCH INFORMATION SYSTEM

- Developed and led by **Ibict**; officially established in 2014.
- Integrates, retrieves, certifies, and visualizes information about Brazil's research ecosystem.
- Organizes fragmented records into a unified national model aligned with international standards.
- Provides free access for researchers, managers, funders, journalists, and citizens.

# 15.9M

publications indexed

Platform count displayed in July 2026

A public gateway to understand who produces knowledge, where, with whom, and with what results.

Source: [brcris.ibict.br](https://brcris.ibict.br) and official BrCris About page, accessed 28 Aug. 2026.



# BrCris connects actors, outputs, and infrastructures

## RESEARCH ACTORS

- Researchers
- Institutions
- Research groups
- Graduate programs
- Funders and projects

## RESEARCH OUTPUTS

- Publications
- Theses and dissertations
- Scientific datasets
- Patents
- Software

## INTEGRATED SOURCES

- Lattes and CNPq directories
- Oasisbr and Capes/Sucupira
- OpenAlex and OpenAIRE
- DOAJ, INPI, and Espacenet
- NDLTLD, ROR, and Wikidata

Source: official BrCris About page and FAQ, accessed 28 Aug. 2026.



# Integration turns records into policy intelligence

## INFORMATION PIPELINE

1. Collect records from national and international sources.
2. Standardize, normalize, and deduplicate the metadata.
3. Connect people, organizations, projects, and results.
4. Provide search, individual profiles, and indicator dashboards.

## DECISIONS IT CAN SUPPORT

- Map collaboration networks and scientific communities.
- Compare institutional and thematic research profiles.
- Identify strengths, gaps, and emerging trends.
- Improve transparency, visibility, and research assessment.
- Ground science, technology, and innovation policies in evidence.

Source: official BrCris About page, Dashboards, and FAQ, accessed 28 Aug. 2026.

# Oasisbr is Brazil's open research gateway

## ONE SEARCH ACROSS OPEN SOURCES

- A multidisciplinary discovery service developed by Ibict.
- Aggregates open content from journals, publication repositories, research data repositories, and digital libraries.
- Provides free access through one interface while directing users to the full object at the source institution.
- Exchanges content with Portuguese infrastructure through RCAAP.

## CONTENT TYPES

- Journal articles and preprints
- Books and book chapters
- Conference papers
- Research datasets
- Theses and dissertations
- Undergraduate works and reports

Source: Oasisbr official portal, About, and FAQ, accessed 28 Aug. 2026.

# BDTD gives visibility to graduate research

## A NATIONAL SERVICE SINCE 2002

- Conceived, developed, and maintained by Ibict.
- Integrates theses and dissertations from Brazilian teaching and research institutions.
- Also includes records of works defended by Brazilians at institutions abroad.
- Offers free national discovery; full texts remain hosted by participating institutions.

## PUBLIC VALUE

- Visibility for postgraduate research
- Shared metadata standards
- Institutional interoperability
- Access to specialized knowledge
- Governance of public investment in graduate education

Source: BDTD official About page, FAQ, and portal, accessed 28 Aug. 2026.



# Oasisbr, BDTD, and BrCris play distinct roles

| Service | Primary question                           | Scope                                                                     | Strategic role                                                    |
|---------|--------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------|
| Oasisbr | What open research outputs can I discover? | Publications, data, books, conference works, theses, and more             | Multidisciplinary access and national/international aggregation   |
| BDTD    | What graduate research has been produced?  | Brazilian theses and dissertations, including records from abroad         | Visibility and governance of postgraduate knowledge               |
| BrCris  | How is the research ecosystem connected?   | Actors, organizations, projects, programs, outputs, patents, and software | Research intelligence, indicators, governance, and policy support |

Together they connect **discovery**, **specialized graduate knowledge**, and **system-level intelligence**.

Sources: official Oasisbr, BDTD, and BrCris portals, accessed 28 Aug. 2026.



# dARK turns persistence into regional sovereignty

## WHAT IT ADDS TO ARK

- **Decentralization:** participating institutions share operational responsibility.
- **Traceability:** identifier events remain auditable over time.
- **Resilience:** distributed governance reduces single points of failure.
- **Sovereignty:** regional institutions define policies and priorities.

## HOW IT WORKS WITH OASISBR

1. Repositories expose qualified metadata through open protocols.
2. Oasisbr aggregates and validates the records.
3. dARK assigns persistent identifiers to eligible objects.
4. Users discover content centrally while access remains with the source institution.



# Open infrastructures advance a focused set of SDGs

## SDG 4 – Quality Education

Open scientific knowledge strengthens higher education, research, and information literacy.

## SDG 9 – Innovation and Infrastructure

Robust, interoperable digital systems provide the backbone for scientific innovation.

## SDG 10 – Reduced Inequalities

Open access reduces institutional, regional, and economic asymmetries in knowledge.

## SDG 16 – Strong Institutions

Transparent information systems support accountability, ethical governance, and evidence-based decisions.

## SDG 17 – Partnerships

Shared standards and interoperable networks enable national and international co-operation.

**Cross-cutting effect:** preservation and responsible reuse also support sustainable cities, institutions, and consumption of digital resources.



# Applied AI creates value when the information base is ready

| Application | AI capability            | Public-interest value                                         |
|-------------|--------------------------|---------------------------------------------------------------|
| ServicesPI  | Semantic search + NLP    | Natural-language discovery across complex patent collections  |
| Patente.IA  | Text + image similarity  | Broader detection of related inventions and prior art         |
| RDAPP       | NLP + semantic retrieval | Structured access and analysis across public-policy documents |
| SAVITS      | BI + ML + LLM            | Monitoring, prediction, and explainable impact assessment     |

The recurring architecture is simple: **infrastructure** → **structured data** → **AI application**.



# Brazil needs a national implementation plan

## POLICY ARCHITECTURE

- A national policy with clear public-interest goals
- Defined roles for MCTI, funders, institutions, and scientific societies
- Shared standards with room for regional and disciplinary adaptation
- Stable funding for open infrastructures and professional teams

## DELIVERY MECHANISMS

- Phased mandates linked to practical support
- Incentives aligned with responsible assessment
- Training, monitoring, and transparent indicators
- Continuous consultation with researchers and society

The objective is gradual, consistent change that increases the public return on research investment.

# The agenda now shifts from isolated projects to durable policy

## SYSTEM PRIORITIES

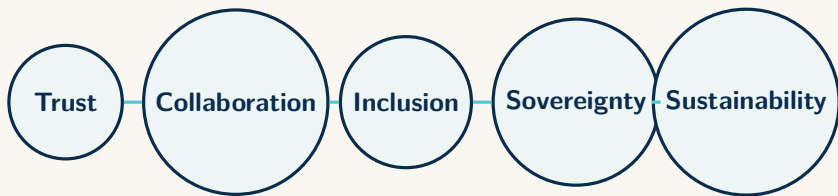
- Sustain public-interest infrastructures.
- Integrate platforms, standards, and preservation.
- Train researchers, librarians, data stewards, and developers.
- Reform incentives and research assessment.
- Govern quality, privacy, bias, and accountability.

## RESEARCHER ACTIONS

- Publish through open routes.
- Share documented data and code.
- Use persistent identifiers and open licenses.
- Join transparent review and citizen science.
- Plan semantics, governance, and preservation from the start.



# Open science is a strategic investment



**Infrastructure + public policy + scientific practice + social movement**

Together, they make research more reliable, reusable, inclusive, and aligned with the public interest.

# License and reuse

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# Obrigado!

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Hindi

Ngiyabonga

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Chinese

Thank you

English

